

CHAPTER 13

Anatomy of the Nervous System

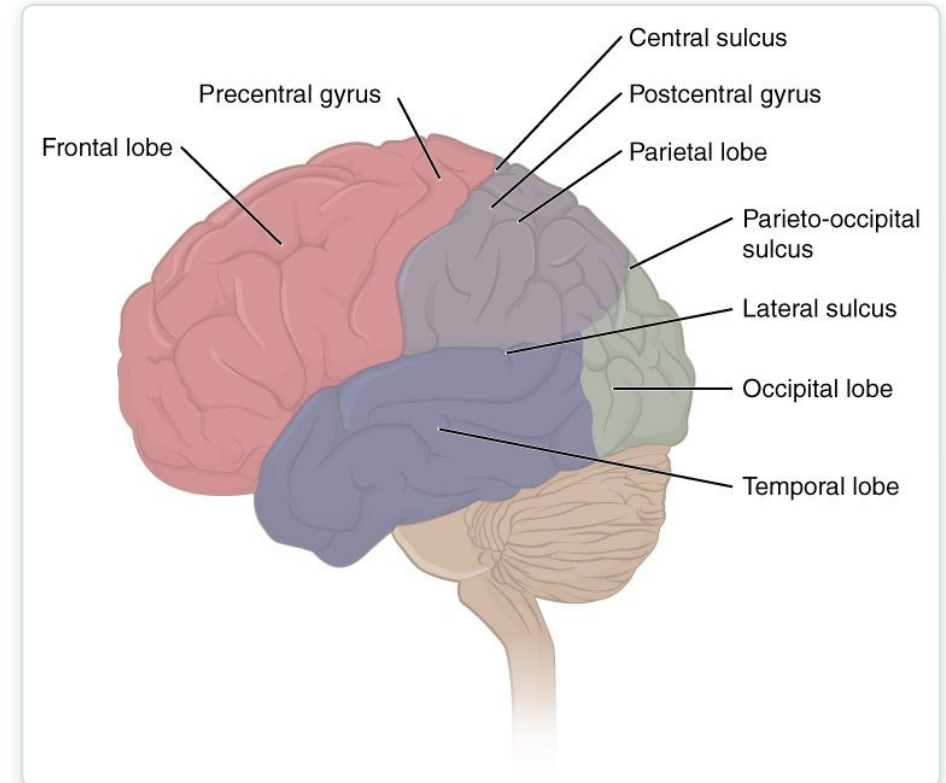
The brain, spinal cord, their coverings, and the peripheral nerves.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Relate the developmental processes of the embryonic nervous system to the adult structures
- 2 Name the major regions of the adult nervous system
- 3 Locate regions of the cerebral cortex on the basis of anatomical landmarks common to all human brains
- 4 Describe the regions of the spinal cord in cross-section
- 5 List the cranial nerves in order of anatomical location and provide the central and peripheral connections



The lobes and landmarks of the cerebrum.



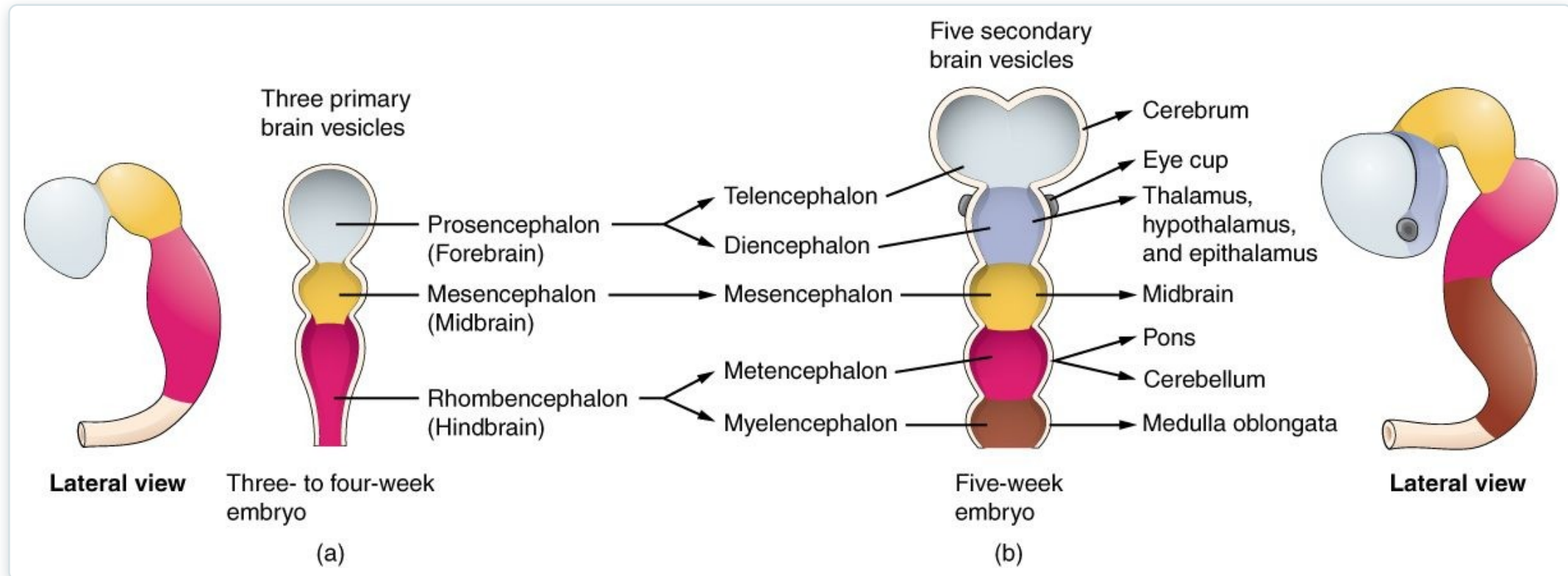
13.1

SECTION

The Embryologic Perspective

How the brain and spinal cord develop from the embryonic neural tube.

Development of the Brain



Brain vesicles and the developing CNS

- The CNS begins as a neural tube
- Its front end forms brain vesicles
- These become forebrain, midbrain, hindbrain
- The rest becomes the spinal cord



13.2

SECTION

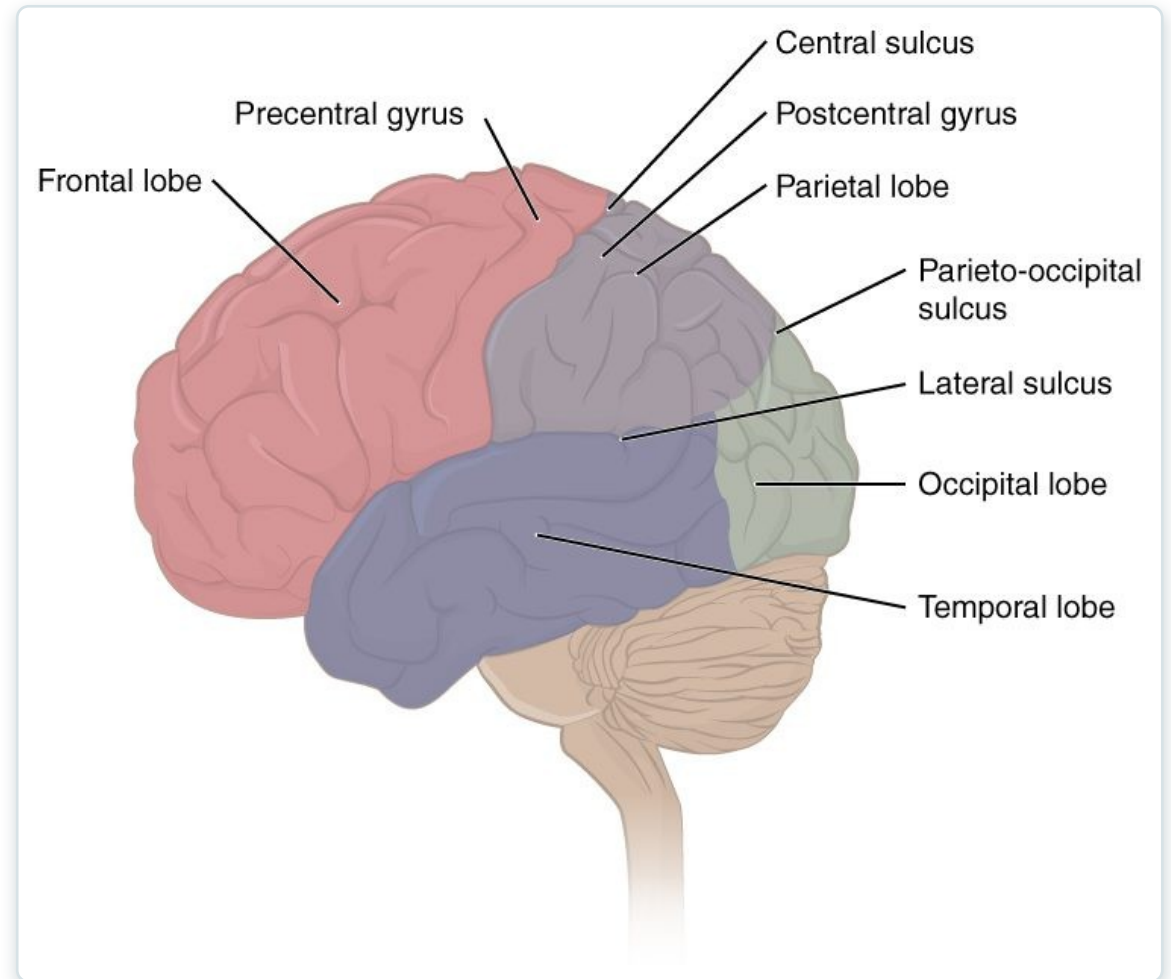
The Central Nervous System

The brain and spinal cord — the body's control and integration center.

SECTION 13.2

Lobes of the Cerebrum

- The largest part of the brain
- Four lobes per hemisphere
- Frontal, parietal, temporal, occipital
- Cortex handles higher functions

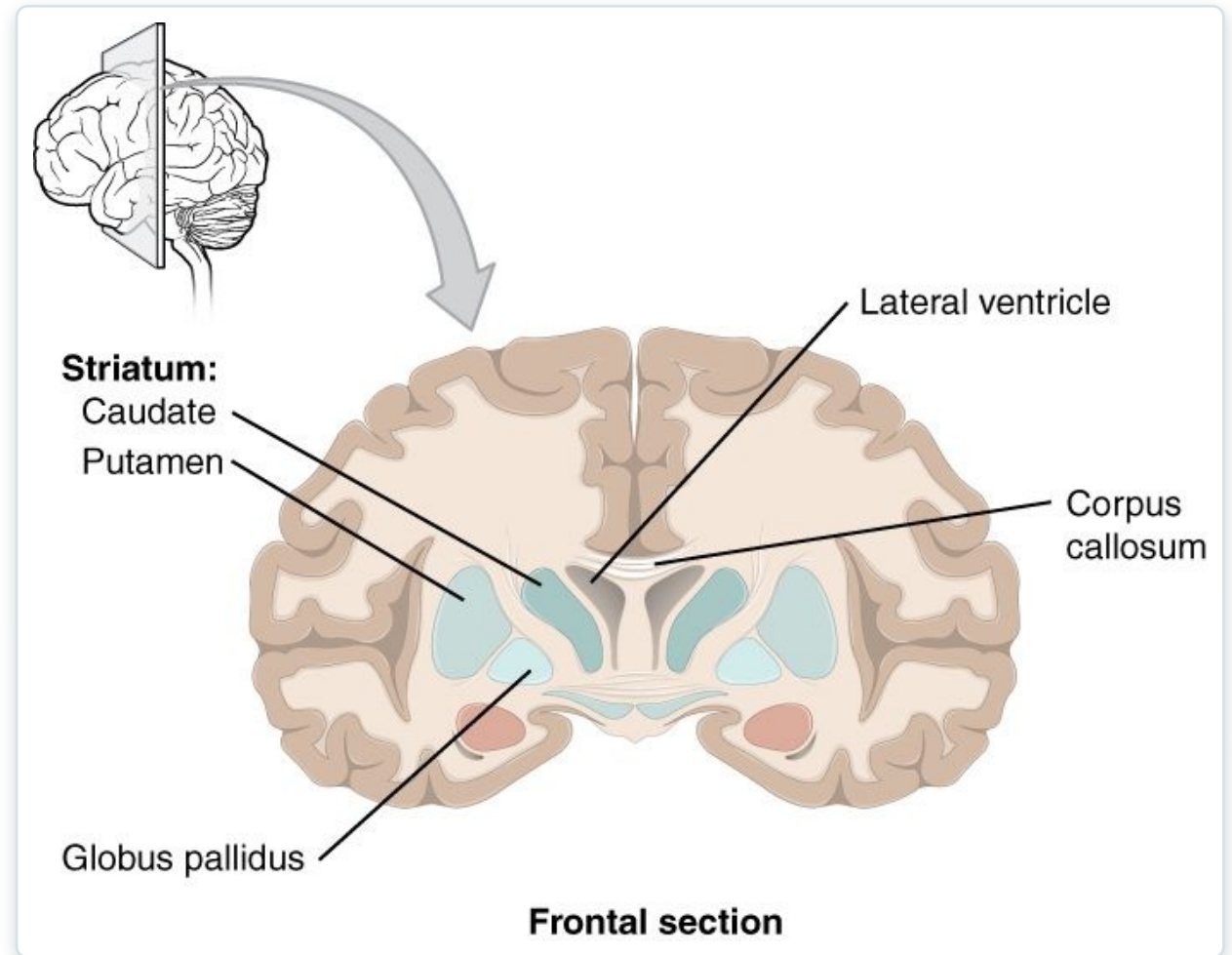


The lobes of the cerebrum

SECTION 13.2

The Basal Ganglia

- Deep gray-matter nuclei
- Help control movement
- Regulate posture and muscle tone
- Affected in Parkinson's disease

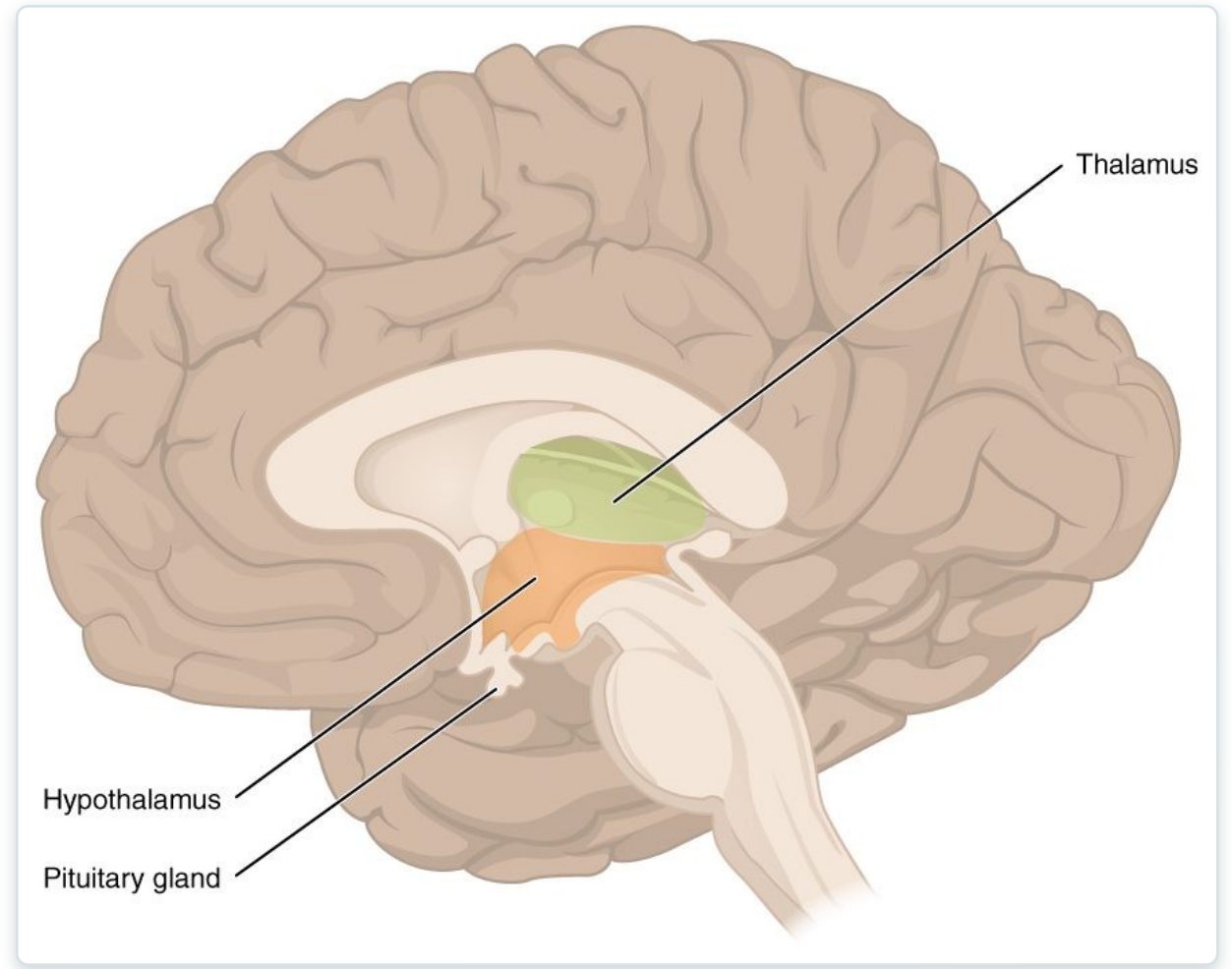


The basal ganglia (frontal section)

SECTION 13.2

The Diencephalon

- Sits between cerebrum and brainstem
- Thalamus relays sensory signals
- Hypothalamus controls homeostasis
- Links to the pituitary gland

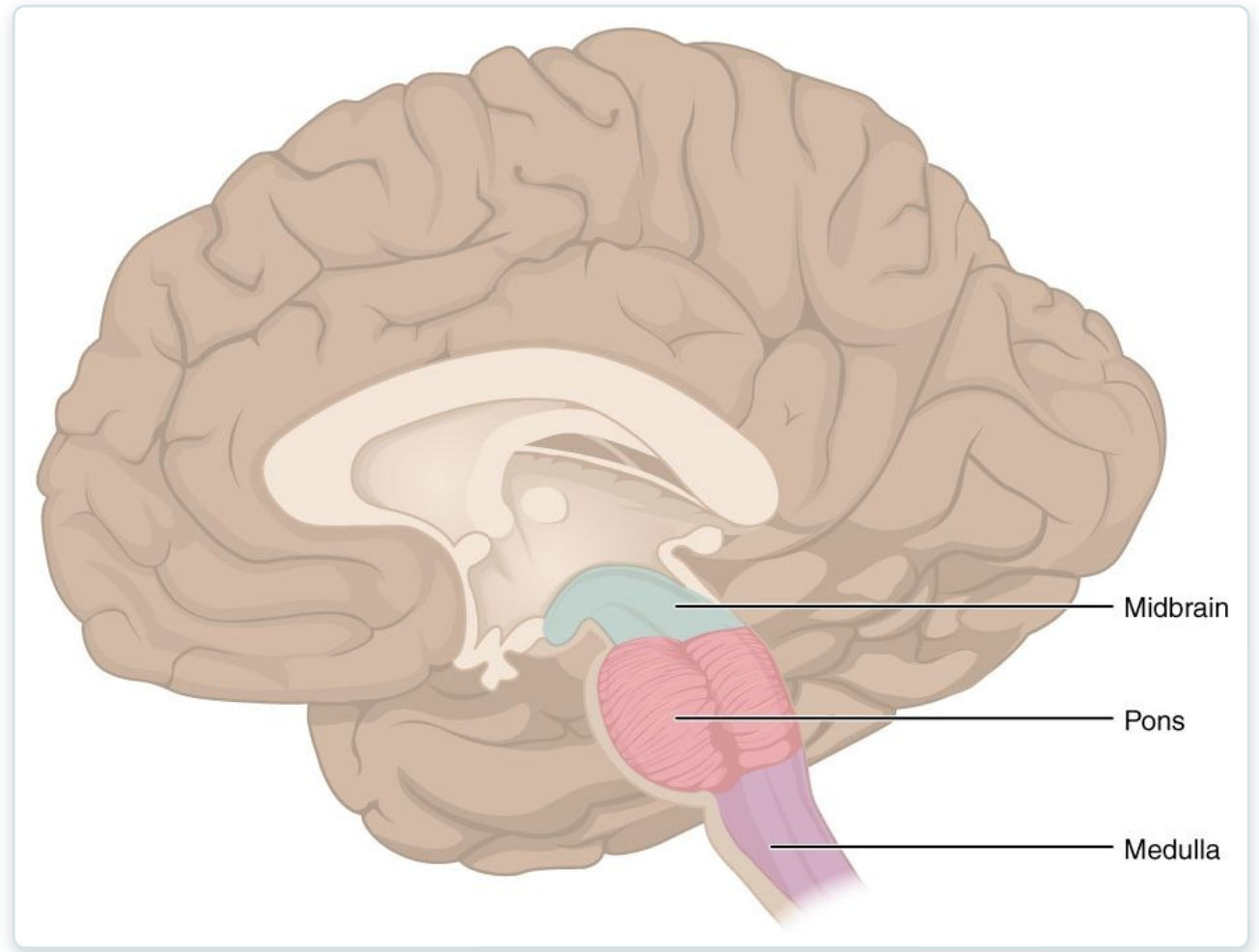


The diencephalon: thalamus and hypothalamus

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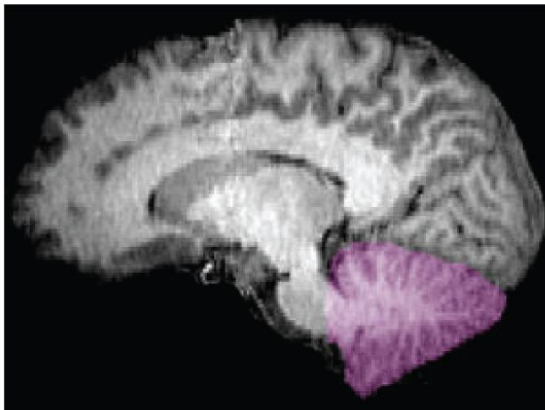
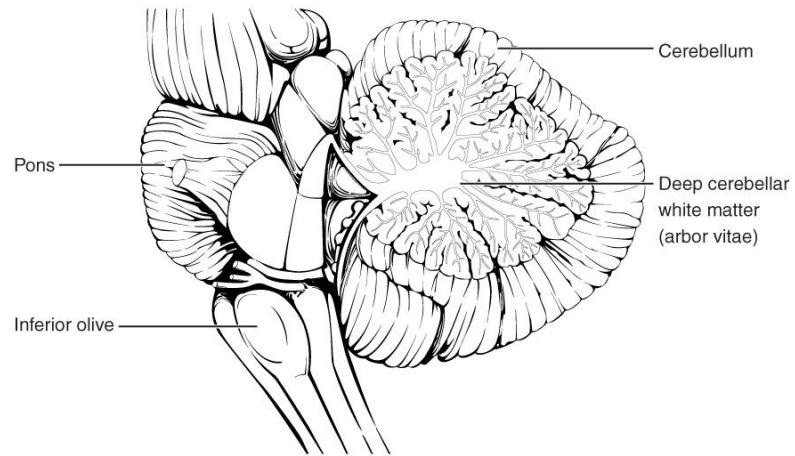
The Brainstem

- Connects brain to spinal cord
- Midbrain, pons, and medulla
- Controls vital automatic functions
- Passage for nerve tracts



The brainstem: midbrain, pons, medulla

The Cerebellum

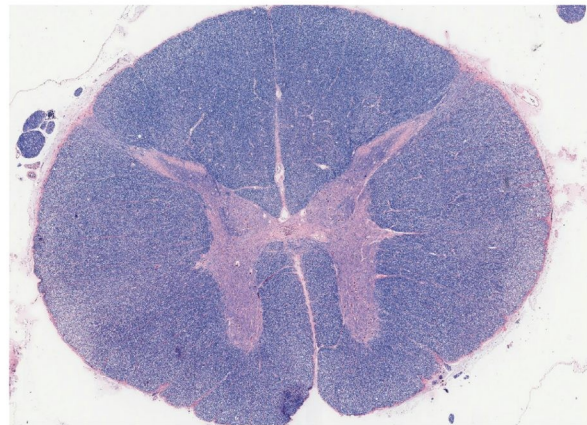
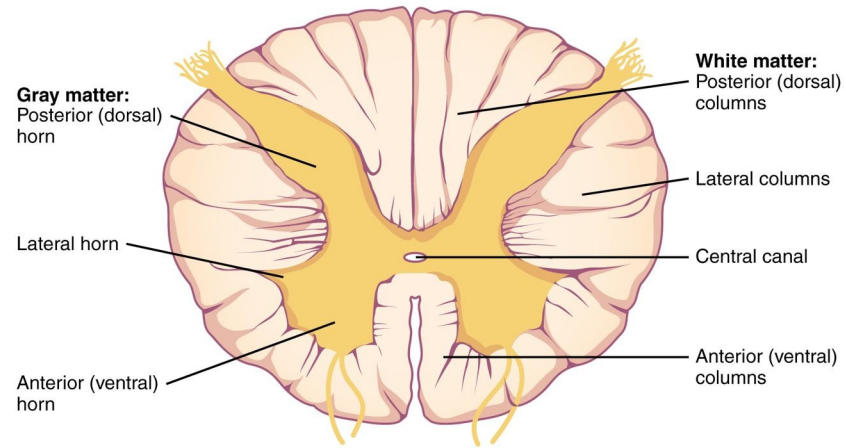


The cerebellum

- Sits behind the brainstem
- Coordinates movement and balance
- Fine-tunes posture and timing
- Enables smooth, accurate motion

SECTION 13.2

The Spinal Cord



Cross-section of the spinal cord

- Continuous with the brainstem
- Gray matter (H-shaped) inside
- White matter tracts outside
- Relays signals and runs reflexes



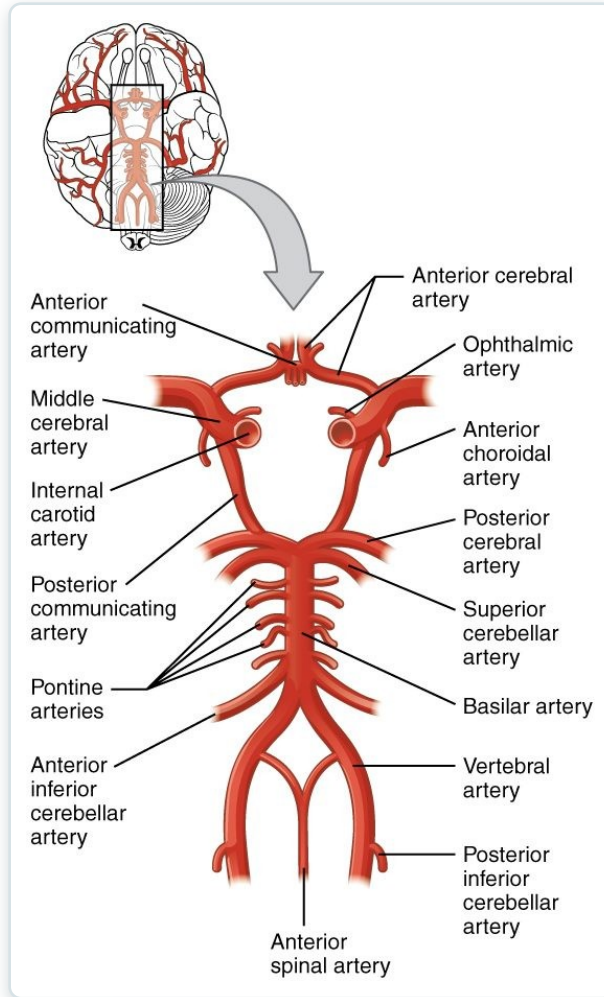
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SECTION

Circulation and the Central Nervous System

Blood supply, the protective meninges, and cerebrospinal fluid.

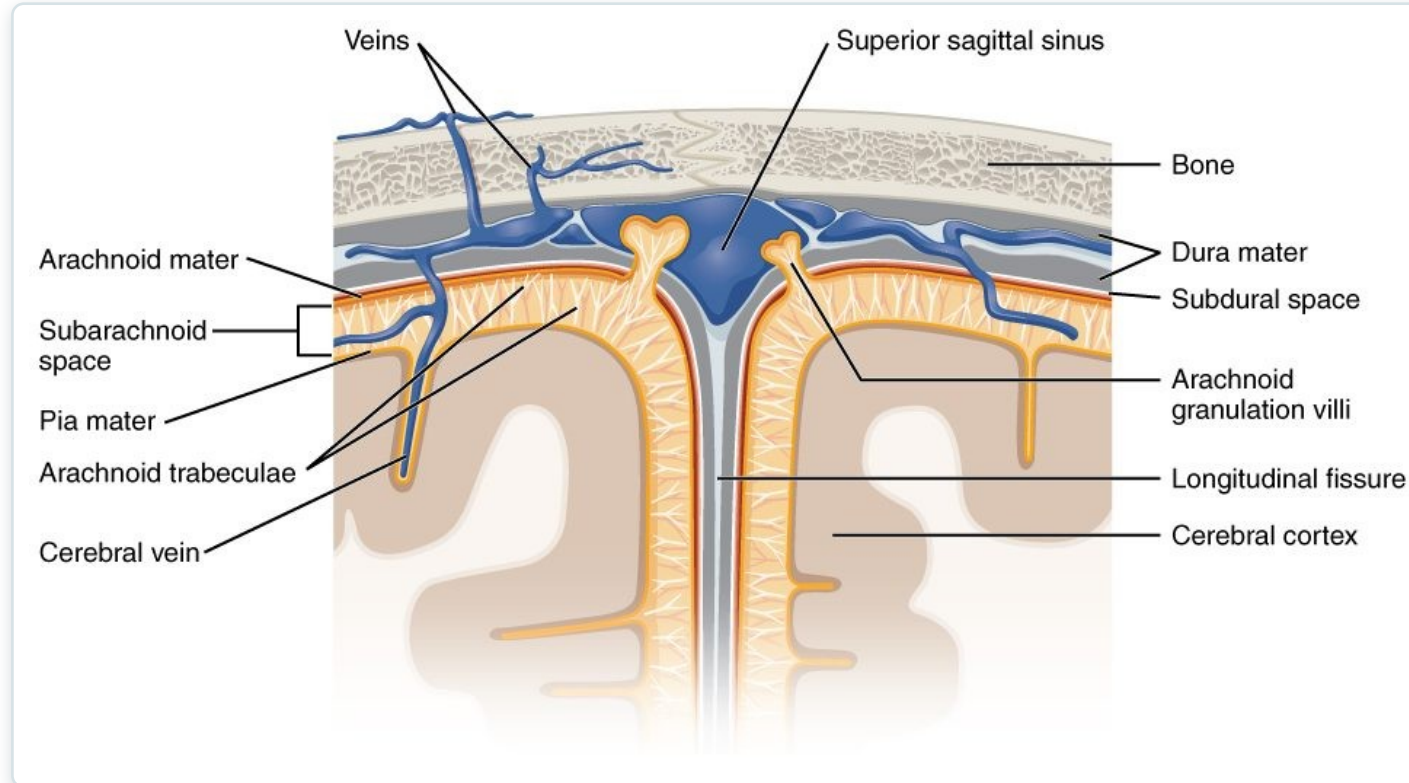
Blood Supply — Circle of Willis



The cerebral arterial circle (Circle of Willis)

- The brain needs constant blood flow
- Fed by carotid and vertebral arteries
- They join in the Circle of Willis
- Provides backup circulation

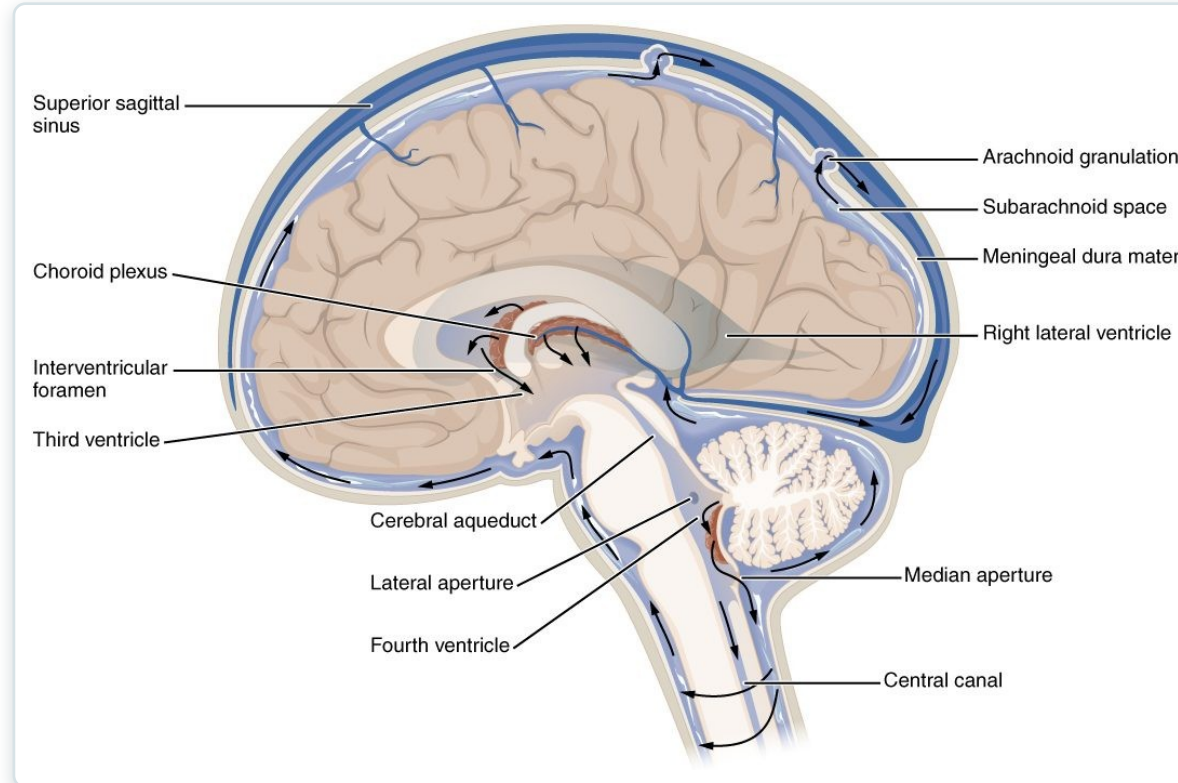
The Meninges



The cranial meninges

- Three protective membranes
- Dura, arachnoid, and pia mater
- Cushion the brain and spinal cord
- Site of meningitis and bleeds

Ventricles & Cerebrospinal Fluid



The ventricles and CSF circulation

- Hollow ventricles inside the brain
- Produce cerebrospinal fluid (CSF)
- CSF cushions and nourishes the CNS
- Circulates and is reabsorbed



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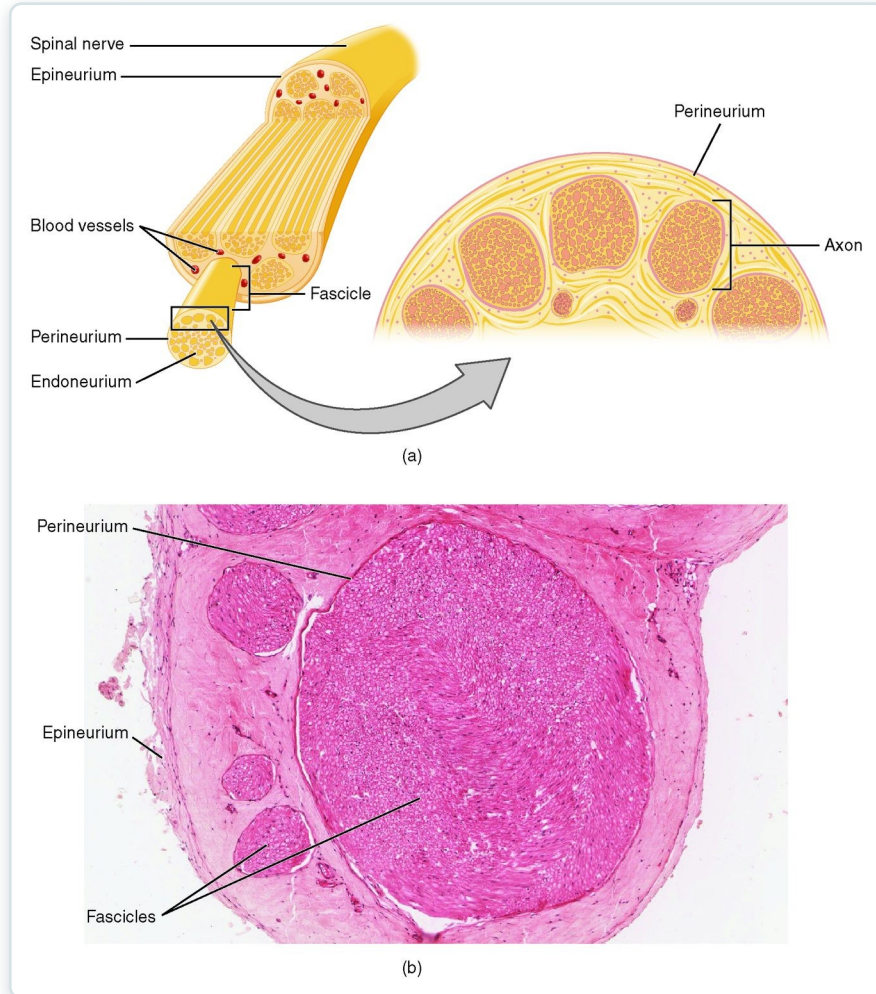
SECTION

The Peripheral Nervous System

The nerves and ganglia that connect the CNS to the body.

SECTION 13.4

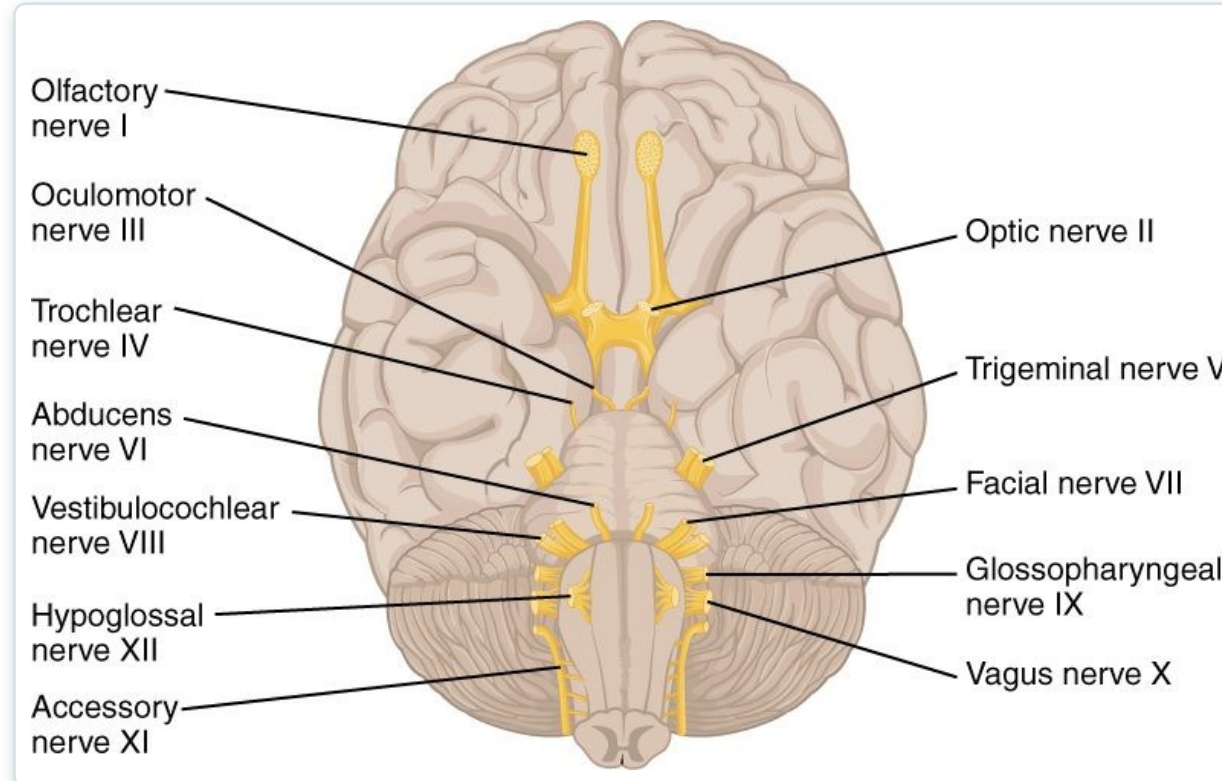
Structure of a Nerve



The structure of a peripheral nerve

- A nerve is a bundle of axons
- Axons grouped into fascicles
- Wrapped in connective-tissue sheaths
- Carry sensory and/or motor fibers

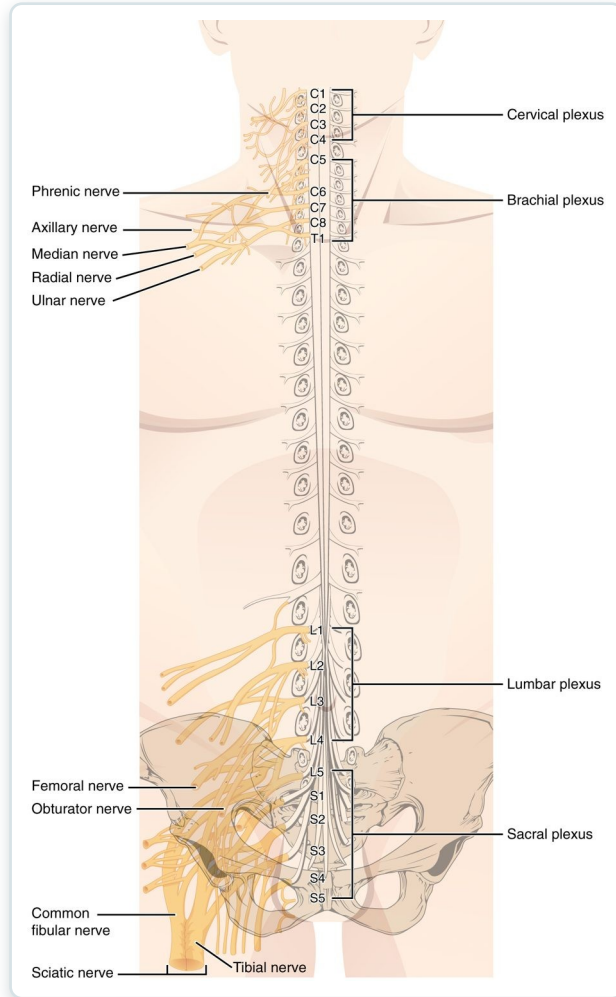
The Cranial Nerves



The twelve cranial nerves

- 12 pairs arise from the brain
- Numbered I through XII
- Serve mainly the head and neck
- Sensory, motor, or both

Spinal Nerves & Plexuses



The spinal nerves and plexuses

- 31 pairs of spinal nerves
- Exit between the vertebrae
- Mixed sensory and motor fibers
- Merge into plexuses for the limbs

Key Takeaways

- The brain and spinal cord develop from the embryonic neural tube.
- The cerebrum, diencephalon, brainstem, and cerebellum each have distinct roles.
- The spinal cord relays signals and runs reflexes, with gray matter inside and white matter outside.
- The brain is protected by meninges, cushioned by CSF, and supplied by the Circle of Willis.
- The PNS connects the CNS to the body through 12 cranial and 31 spinal nerve pairs.

CLINICAL CONNECTIONS

Why Nervous System Anatomy Matters in Practical Nursing

Stroke

Interrupted brain blood flow causes rapid, lasting damage.

Meningitis

Infection of the meninges is a medical emergency.

Spinal Cord Injury

Damage causes loss of function below the level of injury.

Neuro Exam

Cranial-nerve and reflex testing helps localize problems.

Hydrocephalus

Blocked CSF flow raises pressure inside the skull.

Parkinson's Disease

Basal ganglia degeneration impairs movement.